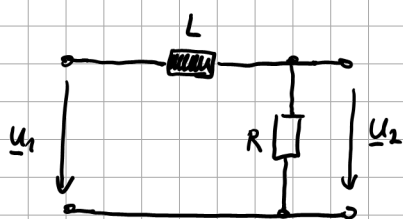


8.5 Tiefpass und Hochpass mit R und L



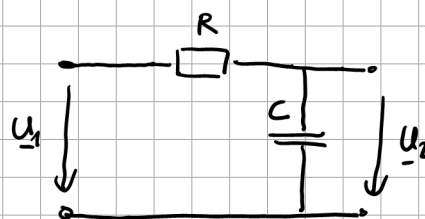
Tiefpass!

$$\underline{A}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = \frac{R}{R + j\omega L} \cdot \frac{\frac{1}{R}}{\frac{1}{R}} = \frac{1}{1 + j\omega \frac{L}{R}} \quad \text{Zeitkonstante}$$

$$NR: \left[\frac{L}{R} \right] = \frac{Vs}{\frac{V}{A}} = s$$

$$|A| = \left| \frac{1}{1 + j\omega \frac{L}{R}} \right| = \frac{1}{\sqrt{1 + (\omega \frac{L}{R})^2}}$$

$$\varphi_A = \varphi_Z - \varphi_N = 0^\circ - \arctan\left(\omega \frac{L}{R}\right)$$



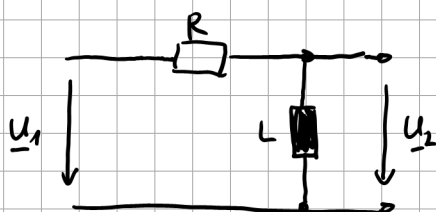
Tiefpass

$$\underline{A}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + R} \cdot \frac{j\omega C}{j\omega C} = \frac{1}{1 + j\omega RC} \quad \text{Zeitkonst.}$$

$$NR: [R \cdot C] = \frac{V}{A} \cdot \frac{As}{V} = s$$

$$|A| = \left| \frac{1}{1 + j\omega RC} \right| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\varphi_A = \varphi_Z - \varphi_N = 0^\circ - \arctan(\omega RC)$$

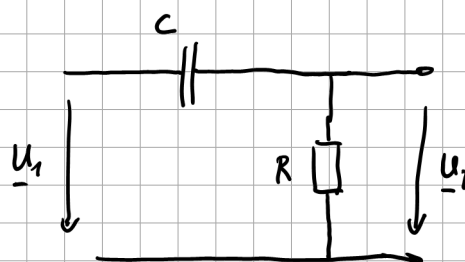


Hochpass

$$\underline{A}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = \frac{j\omega L}{j\omega L + R} \cdot \frac{\frac{1}{R}}{\frac{1}{R}} = \frac{j\omega \frac{L}{R}}{1 + j\omega \frac{L}{R}}$$

$$|A| = \left| \frac{j\omega \frac{L}{R}}{1 + j\omega \frac{L}{R}} \right| = \frac{\omega \frac{L}{R}}{\sqrt{1 + (\omega \frac{L}{R})^2}}$$

$$\varphi_A = \varphi_Z - \varphi_N = 90^\circ - \arctan\left(\omega \frac{L}{R}\right)$$



Hochpass

$$\underline{A}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega RC}{1 + j\omega RC}$$

$$|A| = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

$$\varphi_A = 90^\circ - \arctan(\omega RC)$$

Rechnen mit logarithmischen Zahlenverhältnissen

Amplitudenmaß

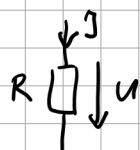
$$a_{dB} = 20 \cdot \log_{10} \left| \frac{U_2}{U_1} \right|$$

Herkunft aus Leistungsdefinition

$$P_{dB} = 10 \cdot \log_{10} \left| \frac{P_2}{P_1} \right|$$

← Messgröße

← Referenz



$$P = \frac{U^2}{R}$$

Einsetzen ergibt

$$P_{dB} = 10 \cdot \log \frac{\left(\frac{U_2^2}{R} \right)}{\left(\frac{U_1^2}{R} \right)} = 10 \cdot \log_{10} \left(\frac{U_2}{U_1} \right)^2$$

Mathe: $\log a^b = b \cdot \log a$

$$= \underbrace{2 \cdot 10}_{20} \cdot \log_{10} \left(\frac{U_2}{U_1} \right)$$

$ A $	a_{dB}
10^3	+ 60 dB
10^2	+ 40 dB
10^1	+ 20 dB
$\sqrt{2}$	+ 3 dB
1	0 dB
$\frac{1}{\sqrt{2}}$	- 3 dB

$ A $	a_{dB}
10^3	+ 60 dB
10^2	+ 40 dB
10^1	+ 20 dB
$\sqrt{2}$	+ 3 dB
1	0 dB
$\frac{1}{\sqrt{2}}$	- 3 dB
10^{-1}	- 20 dB
10^{-2}	- 40 dB
10^{-3}	- 60 dB

Bsp:

$$20 \cdot \log_{10} \left(\frac{1}{\sqrt{2}} \right)$$

$$= 20 \cdot \log_{10} (\sqrt{2})^{-1}$$

$$= - 20 \cdot \log_{10} (\sqrt{2})$$

3 dB

$$= - 3 \text{ dB}$$

Alter in dB, Bsp: 21 Jahre

$$\text{alter}_{dB} = 20 \log_{10} |21| = 26,4 \text{ dB}$$

$$\text{alter-prof}_{dB} = +34,8 \text{ dB}$$

$$20 \log_{10}(x) = 34,8$$

$$10^x \mid \log_{10}(x) = \frac{34,8}{20} \mid 10^x$$

$$x = 10^{\frac{34,8}{20}} = 55$$

Beispiele zum Rechnen

$$|A| = 400$$

$$a_{dB} = ?$$

Mathe $\log(a \cdot b) = \log a + \log b$

$$400 = 4 \cdot 100$$

$$= 2 \cdot 2 \cdot 100$$

$$\Rightarrow a_{dB} = 20 \log |400|$$

$$= 20 \log |2| + 20 \log |2|$$

$$+ 20 \log |100|$$

$$= 20 \log (\sqrt{2})^2 + 20 \log (\sqrt{2})^2$$

$$+ 20 \log 100$$

$$= 2 \cdot 3 \text{ dB} + 2 \cdot 3 \text{ dB} + 40 \text{ dB}$$

$$= 52 \text{ dB}$$

Mathe: $\log \frac{a}{b} = \log a - \log b$

$$|A| = 500 = \frac{1000}{2} = \frac{10^3}{\sqrt{2}^2}$$

$$a_{dB} = 20 \cdot \log(500) = 20 \cdot \log 10^3 - 20 \log (\sqrt{2})^2$$

$$= +60 \text{ dB} - 2 \cdot (20 \log \sqrt{2})$$

$$= 60 \text{ dB} - 2 \cdot 3 \text{ dB}$$

$$= 54 \text{ dB}$$

Matr.-Nr.: $12345 \approx 12000 = 3 \cdot 4 \cdot 1000$

$$3 \approx \sqrt{2}^3$$

$$= \sqrt{2}^3 \cdot 2^2 \cdot 10^3$$

$$\text{matrikel}_{dB} = 20 \cdot \log (\sqrt{2}^3 \cdot \sqrt{2}^4 \cdot 10^3)$$

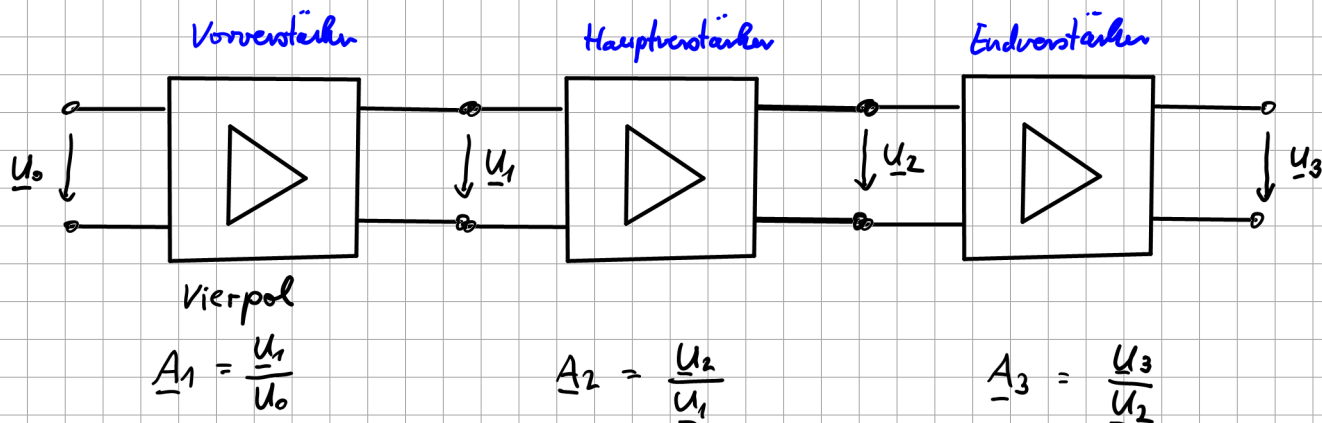
$$= 3 \cdot (20 \log \sqrt{2}) + 4 \cdot (20 \log \sqrt{2}) + 20 \log 10^3$$

$$= 3 \cdot 3 \text{ dB} + 4 \cdot 3 \text{ dB} + 60 \text{ dB}$$

$$= 81 \text{ dB}$$

Taschenrechner: $20 \cdot \log 12345 = 81,8 \text{ dB}$

9 Frequenzgang zusammengesetzter Vierpole



$$\underline{A}_{ges} = \frac{\underline{U}_3}{\underline{U}_0} = \underbrace{\frac{\underline{U}_3}{\underline{U}_2}}_{\underline{A}_3} \cdot \underbrace{\frac{\underline{U}_2}{\underline{U}_1}}_{\underline{A}_2} \cdot \underbrace{\frac{\underline{U}_1}{\underline{U}_0}}_{\underline{A}_1}$$

Multiplikation

$$a_{dB} = 20 \cdot \log_{10} |\underline{A}_{ges}| = 20 \log_{10} |\underline{A}_1 \cdot \underline{A}_2 \cdot \underline{A}_3|$$

$$= 20 \cdot \log |\underline{A}_1| + 20 \log_{10} |\underline{A}_2| + 20 \log_{10} |\underline{A}_3|$$

$$a_{dB} = a_{1dB} + a_{2dB} + a_{3dB}$$

$$\angle \underline{A} = \angle (\underline{A}_1 \cdot \underline{A}_2 \cdot \underline{A}_3)$$

$$= \angle (\underline{A}_1) + \angle (\underline{A}_2) + \angle (\underline{A}_3)$$

$$\varphi_{ges} = \varphi_1 + \varphi_2 + \varphi_3$$